

Supplement: From Accuracy to Allocation

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1 Extended Notes on Data Provenance

Sources and licenses.

- **New York:** *311 Service Requests from 2010 to Present*, NYC Open Data, dataset `erm2-nwe9`; NYC Open Data terms (free use without restriction).
- **Chicago:** *311 Service Requests*, Chicago Data Portal, dataset `v6vf-nfxy`; covers the current 311 system live since 2018-12-18 (the study window starts 2019-01-01 to avoid the system transition); source-flagged duplicates dropped.
- **San Francisco:** *311 Cases*, DataSF, dataset `vw6y-z8j6`; Public Domain Dedication and License.
- **Austin:** *Austin 311 Public Data*, City of Austin Open Data Portal, dataset `i26j-ai4z`.
- **Weather:** NOAA GHCN-Daily through the AWS Open Data Program mirror `noaa-ghcn-pds` (maintained by NOAA NCEI); stations USW00094728 (Central Park), USW00094846 (O'Hare), USW00023272 (San Francisco Downtown), USW00013904 (Camp Mabry). Native units (tenths of mm/°C, tenths of m/s) converted only in preprocessing.

Acquisition queries. Each 311 source is queried through the SODA API with server-side aggregation — `$select=date_trunc_ymd(created_date), category, count(*)`, grouped and ordered, chunked by calendar year, paginated at 50,000 rows — so the rawest stored layer is already free of record-level attributes. The complete list of issued query URLs for every city is in `data/raw/311/*manifest.json`.

Known data characteristics. Reported request volumes reflect intake-channel structure (phone/app/web), administrative practice, and reporting propensity, all of which differ across cities and over time; the Chicago information-only and aircraft-noise exclusions and the SF duplicate exclusion are documented with reasons in `configs/service_families.yml`. A missing (day, family) pair within a city’s observed range denotes zero requests created, and the panel densification step materializes those zeros explicitly.

2 Service-Family Harmonization

The harmonization rule set (`configs/service_families.yml`) is an ordered, case-insensitive substring matcher: the first matching family wins, and categories matching no rule fall to *other*. Order resolves genuine ambiguities deterministically (e.g., “noise” complaints about construction are noise, not housing/building, because the noise rule precedes). The full mapping from every native category to its family, with request counts, is exported per city to `outputs/tables/family_composition_{city}.csv` by the preprocessing stage and ships with the repository; the tables below summarize the dominant native categories per family in each city.

Two properties matter for interpretation. First, the rules are *versioned*: any change to the mapping is a visible diff, and the unit test suite pins representative mappings. Second, family compositions are *not* identical across cities — e.g., the noise family is dominated by residential-noise complaints in New York but is a much smaller share elsewhere; the sanitation family includes encampment-related categories in San Francisco that have no Chicago counterpart. All cross-city statements in the main text are therefore about families as *planning buckets within each city*, and the cross-city axis compares the structure of the accuracy-to-decision mapping, never raw family volumes.

[PENDING-REAL-RESULTS: $s_{harmonization_tables}$]

3 Model Configurations

Hyperparameters are fixed across cities and feature sets so that no configuration receives a larger tuning budget than another; values were chosen a priori in the ranges standard for daily-count panels of this size and are not per-task optimized.

Model	Parameters	Notes
Trailing mean	7-day window	forecast-free baseline
Seasonal naive	same weekday, $t - 6$	forecast-free baseline
Ridge	$\alpha = 1.0$, standardized inputs	
Random forest	300 trees, min leaf 5, 60% features	Breiman (2001)
LightGBM point	L1 objective, 600 trees, lr 0.05, 63 leaves, min child 30	Ke et al. (2017)
LightGBM quantile	as above, pinball objective per level non-crossing via cumulative max	levels .05/.25/.50/.75/.95

Categorical handling: one-hot family indicators everywhere; one-hot city indicators in pooled (global) models only. Predictions are clipped at zero. The selected configuration per (scope, city, feature set, model) is refit on train+validation before its single test evaluation, exactly as in the protocol of the main text.

4 Allocation Optimality

The greedy allocation used by all expected-value policies is exactly optimal for the per-day problem it solves.

Proposition 1. *Let $D_1, \dots, D_S$ be non-negative random variables, $\omega_s > 0$, $\kappa > 0$, and consider*

$$\max_{x \in \mathbb{Z}_{\geq 0}^S, \sum_s x_s = B} \sum_{s=1}^S \omega_s \mathbb{E}[\min(D_s, \kappa x_s)].$$

The greedy algorithm that assigns crews one at a time to the family with the largest marginal gain attains the optimum.

Proof. For fixed s , define $g_s(x) = \omega_s \mathbb{E}[\min(D_s, \kappa x)]$ on integers $x \geq 0$. For any realization $d \geq 0$ the map $c \mapsto \min(d, c)$ is concave in c ; expectation preserves concavity, and restriction to the integer grid yields non-increasing increments $\Delta_s(x) = g_s(x+1) - g_s(x)$. The problem is therefore the maximization of a separable discrete-concave function over the simplex $\sum_s x_s = B$, for which the greedy/incremental algorithm is optimal: at every step the chosen increment is at least as large as any increment available later (by monotonicity of Δ_s), so an exchange argument converts any optimal solution into the greedy one without loss of objective value. $\square$

In the implementation $\mathbb{E}[\min(D_s, c)]$ is evaluated exactly for the discrete distribution defined by linear interpolation of the inverse CDF through the five forecast quantiles on a fixed 99-point grid (src/optimization/allocation.py); point forecasts use a degenerate distribution, and the oracle uses the realized value. The unit test `test_greedy_matches_bruteforce_on_small_instance` verifies greedy-vs-brute-force equality on random instances.

5 Full Result Tables

[PENDING-REAL-RESULTS: `s_tables`]

6 Negative and Conditional Results

[PENDING-REAL-RESULTS: $s_{negative}$]

7 Sensitivity Analyses

[PENDING-REAL-RESULTS: $s_{sensitivity}$]

8 Computational Resources

[PENDING-REAL-RESULTS: $s_{compute}$]

9 Ethics and Responsible-AI Statement

Data. All inputs are public administrative aggregates (daily counts of service requests by category) and public weather observations. No individual-level, household-level, or address-level information is acquired, stored, or processed; the acquisition layer requests server-side aggregation precisely so that record-level attributes (including free-text descriptions and locations) never enter the pipeline.

People inside the data. Each request count summarizes residents’ attempts to obtain public services. Two ethical asymmetries follow. First, reported volume under-represents communities with lower propensity or ability to report ([Kontokosta et al., 2017](#); [Kontokosta and Hong, 2021](#)); optimizing service against reported volume can therefore divert resources from quiet need toward vocal demand. This study’s city-level action space cannot reallocate across neighbourhoods, which contains but does not eliminate the concern: the family-level analogue (e.g., systematically deprioritizing housing complaints relative to noise) is measured and reported per policy. Second, the priority weights are normative inputs; we publish them as configuration, sweep them, and explicitly assign their real-world selection to accountable human decision-makers.

Intended and unintended use. The system evaluated here is a research benchmark. It is not a deployable dispatch tool, has no city involvement or endorsement, and the manuscript states this in every section where simulation results appear. Misuse risk (treating simulated capacity numbers as operational guidance) is mitigated by classifying all operational parameters as sensitivity-only and reporting ranges rather than recommendations.

Human oversight. The intended human-AI division of labour — forecasts and allocation suggestions with uncertainty made visible, final allocation authority retained by planners — is described in the discussion and is a design commitment of the broader research programme, not an evaluated capability of this study.

10 Reproducibility Instructions

One-command rebuild. From the repository root:

```

pip install -r requirements.txt
make test          # 12+ unit tests: harmonization determinism,
                   # leakage controls, split chronology,
                   # allocation optimality, simulator conservation
make all           # panel -> features -> forecasts -> decisions
                   # -> all tables and figures in outputs/

```

Determinism. A single global seed (20260609) controls every stochastic component (NumPy, scikit-learn, LightGBM); expected-value allocation uses a deterministic 99-point inverse-CDF grid rather than Monte Carlo, so decision results are bit-stable given forecasts.

Data provenance. Raw acquired layers are versioned in `data/raw/` together with JSON manifests recording: source authority, dataset id and page, license note, every query URL issued, retrieval timestamps (UTC), uncompressed and compressed SHA-256 checksums, row counts, and — for 311 sources — the portal’s own dataset meta-data (`rowsUpdatedAt`, column inventory) at retrieval time. Re-running `make acquire-311` `acquire-weather` re-derives the same layers from the sources (counts for recent periods may drift as portals backfill; the manifests pin what this paper used).

Environment. Python 3.11; exact package versions in `reproducibility/pip-freeze.txt`; no GPU required; total pipeline runtime and per-stage costs in Section 8 of this supplement.

Acquisition environment note. The research sandbox used for this project blocks direct access to the four city portals; the 311 acquisition therefore ran through the repository’s GitHub Actions workflow (`.github/workflows/acquire-311-data.yml`) on GitHub-hosted runners, executing the same committed script and pushing the raw aggregates and manifests back to the branch. The first attempt with full-window queries timed out against the NYC portal and was replaced by per-calendar-year chunking; both attempts are preserved in the repository’s Actions history.

References

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